

CSE 301 — Probability & Statistics

Lecture Notes (Lectures 1–17)

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1 Introduction to Probability

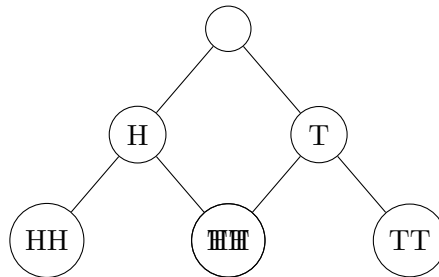
History & Law: Mosteller–Wallace (authorship problem), Fermat, Pascal, Newton.

“Statistics is the logic of uncertainty.”

Sample space S : the set of all possible outcomes of an experiment.

Event: a subset of the sample space.

Example: Flip a coin twice. The sample space is built up as a tree of outcomes:



$$S = \{HH, HT, TH, TT\}$$

Naive definition of probability:

$$P(A) = \frac{\text{\#favorable outcomes}}{\text{\#possible outcomes}}$$

(Assumes all outcomes in S are equally likely.)

Counting — Multiplication Rule: If one experiment has n_1 outcomes, and for each outcome of the 1st experiment there are n_2 possible outcomes for the 2nd experiment, $\dots$, n_r possible outcomes for the r -th experiment, then there are $n_1 n_2 \cdots n_r$ overall possible outcomes.

Example — Probability of a full house in poker (5-card hand):

A full house = three cards of one rank + two cards of another rank.

- Choose rank for the triple: 13 ways; choose 3 of the 4 suits: $\binom{4}{3} = 4$
- Choose rank for the pair: 12 ways; choose 2 of the 4 suits: $\binom{4}{2} = 6$

$$P(\text{full house}) = \frac{13 \binom{4}{3} \cdot 12 \binom{4}{2}}{\binom{52}{5}}$$

2 Counting & Story Proofs

Sampling table — choose k objects out of n :

	Order matters	Order doesn't matter
With replacement	n^k	$\binom{n+k-1}{k}$
Without replacement	$n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$	$\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Equivalent problem: How many ways to put k indistinguishable objects into n distinguishable boxes? This is the “stars and bars” construction — k dots and $n-1$ separators arranged in a row:

$$\begin{array}{ccccccc} \bullet & \bullet & \bullet & | & \bullet & \bullet & | & \bullet & \bullet & \bullet & \bullet \\ \text{box 1} & & & & \text{box 2} & & & \text{box 3} & & & \end{array}$$

$$\Rightarrow \binom{n+k-1}{k}$$

Story proofs:

1. $\binom{n}{k} = \binom{n}{n-k}$
2. $n\binom{n-1}{k-1} = k\binom{n}{k}$ — pick k people out of n and designate one of them captain (count two ways).

3. **Vandermonde's identity:**

$$\binom{m+n}{k} = \sum_{j=0}^k \binom{m}{j} \binom{n}{k-j}$$

Non-naive (axiomatic) definition of probability: A probability space consists of S (sample space) and P , a function that takes an event $A \subseteq S$ as input and gives $P(A) \in [0, 1]$ such that:

1. $P(\emptyset) = 0$, $P(S) = 1$
2. $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$ if the A_i are disjoint.

3 The Birthday Problem & Properties of Probability

Birthday Problem: Given k people, find the probability that two of them share a birthday.

If $k > 365$, then $P = 1$ (pigeonhole principle). Assume birthdays are equally likely and independent. For $k \leq 365$:

$$P(\text{no match}) = \frac{365 \cdot 364 \cdots (365 - k + 1)}{365^k}$$

$$P(\text{match}) = 1 - P(\text{no match})$$

k	$P(\text{match})$
23	$\approx 50.7\%$
50	$\approx 97\%$
100	$\approx 99.9994\%$

Properties of probability:

1. $P(A^c) = 1 - P(A)$

Proof:

$$1 = P(S) = P(A \cup A^c) = P(A) + P(A^c) \Rightarrow P(A^c) = 1 - P(A)$$

2. If $A \subseteq B$, then $P(A) \leq P(B)$.

Proof: $B = A \cup (B \cap A^c)$ is a disjoint union, so

$$P(B) = P(A) + P(B \cap A^c) \geq P(A)$$

3. Inclusion–exclusion (two events):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof:

$$P(A \cup B) = P(A \cup (B \cap A^c)) = P(A) + P(B \cap A^c)$$

It remains to show $P(B) - P(A \cap B) = P(B \cap A^c)$, which follows since

$$P(B) = P(A \cap B) + P(B \cap A^c)$$

4. Disjoint unions / three events:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

5. **General principle of inclusion–exclusion:**

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \cdots$$

De Montmort's Matching Problem (1708): Shuffle cards labeled $1, 2, \dots, n$. Let A_i be the event that the i -th card matches (i.e., card i lands in position i).

4 The Matching Problem & Conditional Probability

Continuing the matching problem:

$$\begin{aligned}P(A_i) &= \frac{1}{n}, \\P(A_i \cap A_j) &= \frac{(n-2)!}{n!}, \\P(A_{i_1} \cap \dots \cap A_{i_k}) &= \frac{(n-k)!}{n!}\end{aligned}$$

By inclusion-exclusion:

$$\begin{aligned}P\left(\bigcup_i A_i\right) &= \binom{n}{1}\frac{1}{n} - \binom{n}{2}\frac{1}{n(n-1)} + \binom{n}{3}\frac{(n-3)!}{n!} - \dots \\&= 1 - \frac{1}{2!} + \frac{1}{3!} - \dots \pm \frac{1}{n!}\end{aligned}$$

As $n \rightarrow \infty$, this approaches

$$P(\text{at least one match}) \rightarrow 1 - e^{-1} \approx 0.632$$

(Aside mentioned in notes: Strassen's algorithm for fast matrix multiplication, $O(n^{2.81})$.)

Conditional Probability — Independence (Definition):

Events A, B are **independent** if $P(A \cap B) = P(A)P(B)$.

Events A, B, C are **independent** if they're pairwise independent

$$P(A, B) = P(A)P(B), \quad P(B, C) = P(B)P(C), \quad P(C, A) = P(C)P(A)$$

and $P(A, B, C) = P(A)P(B)P(C)$.

Newton-Pepys Problem (1693): Which is more likely, when rolling fair dice?

$$P(A) = P(\text{at least one 6 in 6 dice}) = 1 - \left(\frac{5}{6}\right)^6 \approx 0.665$$

$$P(B) = P(\text{at least two 6s in 12 dice}) = 1 - \left(\frac{5}{6}\right)^{12} - 12 \cdot \frac{1}{6} \left(\frac{5}{6}\right)^{11} \approx 0.619$$

$$P(C) = P(\text{at least three 6s in 18 dice}) = 1 - \sum_{k=0}^2 \binom{18}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{18-k} \approx 0.597$$

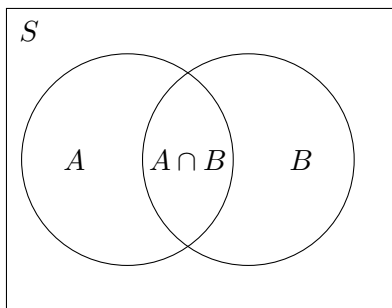
So $P(A) > P(B) > P(C)$ (event A is most likely).

Conditional probability — how we update belief/uncertainty based on new evidence:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

5 Conditional Probability (continued)

Definition: $P(A | B) = \frac{P(A \cap B)}{P(B)}$, for $P(B) > 0$.



Theorem 1: $P(A \cap B) = P(B)P(A | B) = P(A)P(B | A)$

Theorem 2 (Chain Rule):

$$P(A_1, A_2, \dots, A_n) = P(A_1) P(A_2 | A_1) P(A_3 | A_1, A_2) \cdots P(A_n | A_1, \dots, A_{n-1})$$

Theorem 3 (Bayes' Rule):

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Law of Total Probability:

$$P(B) = P(A_1)P(B | A_1) + P(A_2)P(B | A_2) + \cdots + P(A_n)P(B | A_n)$$

Example: Get a random 2-card hand from a standard deck. Compare:

$$P(\text{both aces} | \text{have an ace}) \quad \text{vs} \quad P(\text{both aces} | \text{have the ace of spades})$$

$$P(\text{both aces} | \text{have an ace}) = \frac{P(\text{both aces})}{P(\text{have an ace})} = \frac{\binom{4}{2} / \binom{52}{2}}{1 - \binom{48}{2} / \binom{52}{2}} = \frac{1}{33} \approx 3.03\%$$

$$P(\text{both aces} | \text{have ace of spades}) = \frac{P(\text{both aces})}{P(\text{have ace of spades})} = \frac{1}{17} \approx 5.88\%$$

Example — Disease testing:

A disease affects 1% of the population: $P(D) = 0.01$, $P(D^c) = 0.99$. A test is 95% accurate: $P(+ | D) = 0.95$, $P(+ | D^c) = 0.05$.

By Bayes' rule:

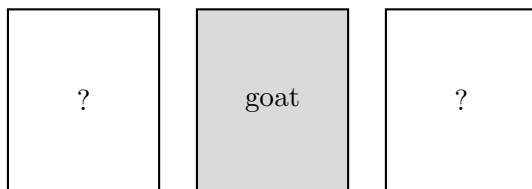
$$\begin{aligned} P(D \mid +) &= \frac{P(+ \mid D) P(D)}{P(+ \mid D) P(D) + P(+ \mid D^c) P(D^c)} \\ &= \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.05 \times 0.99} \\ &\approx 0.16 \end{aligned}$$

6 Monty Hall & Simpson's Paradox

Definition (Conditional Independence): Events A and B are conditionally independent given C if

$$P(A \cap B \mid C) = P(A \mid C) P(B \mid C)$$

Monty Hall Problem: One door has a car, the other two have goats. Monty knows which is which. After the contestant picks a door, Monty opens one of the remaining doors, always revealing a goat. (If he has a choice of two goat doors, he picks uniformly at random.) Should the contestant switch?



Door 1 (your pick) Door 2 (opened: goat) Door 3 (switch?)

Let D_j = “door j has the car,” $j \in \{1, 2, 3\}$, and S = “success, assuming the contestant switches.” Suppose the contestant picks door 1 and Monty opens door 2.

$$\begin{aligned} P(S \mid \text{Monty opens door 2}) &= P(S \cap D_1) + P(S \cap D_2) + P(S \cap D_3) \\ &= P(S \mid D_1)P(D_1) + P(S \mid D_2)P(D_2) + P(S \mid D_3)P(D_3) \\ &= 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} \\ &= \frac{2}{3} \end{aligned}$$

So **switching wins with probability 2/3**, staying wins with probability 1/3.

Simpson's Paradox: (Example: Dr. Hibbert vs. Dr. Nick, comparing surgery success rates.)

	Bandage Procedure	Heart Surgery
Dr. Hibbert	77% success	90% success
Dr. Nick	81% success	0% success

(Despite Dr. Nick having a higher rate in each category, his overall success rate can end up lower, due to differing caseloads — this is the paradox.)

- A : successful surgery
- B : treated by Dr. Nick
- C : heart surgery

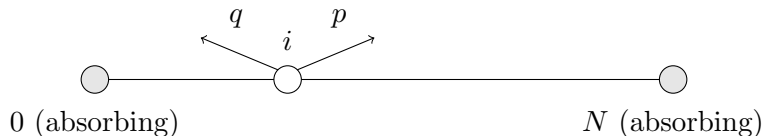
Within each category: $P(A \mid B, C) < P(A \mid B^c, C)$ and $P(A \mid B, C^c) < P(A \mid B^c, C^c)$, but overall: $P(A \mid B) > P(A \mid B^c)$.

This is explained by differing **weights** (case-mix) between categories:

$$P(A \mid B) = P(C \mid B) P(A \mid B, C) + P(C^c \mid B) P(A \mid B, C^c)$$

7 Gambler's Ruin & Random Variables

Gambler's Ruin: Two gamblers A and B play a sequence of \$1 rounds; $p = P(A \text{ wins a round})$, $q = 1 - p$. Find the probability that A wins the entire game (i.e., B goes bankrupt), assuming A starts with \$ i and B starts with \$ $(N - i)$.



Strategy: Condition on the first step. Let $p_i = P(A \text{ wins} \mid A \text{ starts with } \$i)$. Then:

$$p_i = p p_{i+1} + q p_{i-1}, \quad 1 \leq i \leq N-1, \quad p_0 = 0, \quad p_N = 1$$

This is a linear difference equation. Try $p_i = x^i$:

$$p x^2 - x + q = 0 \quad \Rightarrow \quad x = 1 \quad \text{or} \quad x = q/p$$

General solution $p_i = A + B(q/p)^i$. Applying $p_0 = 0$, $p_N = 1$:

$$p_i = \frac{1 - (q/p)^i}{1 - (q/p)^N} \quad (p \neq q)$$

$$p_i = \frac{i}{N} \quad (p = q = \frac{1}{2})$$

Random Variable: A function from the sample space S to $\mathbb{R}$ — think of it as a numerical summary of an aspect of an experiment.

Definition (Bernoulli): A r.v. X has the **Bernoulli distribution**, $X \sim \text{Bern}(p)$, if X takes two possible values 0, 1, with $P(X = 1) = p$, $P(X = 0) = 1 - p = q$.

8 The Binomial Distribution

$\text{Bin}(n, p)$: the distribution of the number of successes X in n independent $\text{Bern}(p)$ trials.

1. **Story:** X = number of successes in n independent $\text{Bern}(p)$ trials.

2. **Sum of indicators:** $X = X_1 + X_2 + \cdots + X_n$, where

$$X_i = \begin{cases} 1 & \text{if the } i\text{-th trial is a success} \\ 0 & \text{otherwise} \end{cases}$$

and $X_1, \dots, X_n$ are i.i.d. $\text{Bern}(p)$.

PMF:

$$P(X = k) = \binom{n}{k} p^k q^{n-k}, \quad q = 1 - p, \quad k = 0, 1, \dots, n$$

CDF: $F(x) = P(X \leq x)$.

Theorem: If $X \sim \text{Bin}(n, p)$, $Y \sim \text{Bin}(m, p)$, independent, then $X + Y \sim \text{Bin}(n + m, p)$.

Proof 1 (story): n trials with X successes, plus m more independent trials with Y successes, together form $n + m$ trials with $X + Y$ successes.

Proof 2 (indicators + convolution):

$$\begin{aligned} P(X + Y = k) &= \sum_{j=0}^k P(X = j)P(Y = k - j) \\ &= \sum_{j=0}^k \binom{n}{j} p^j q^{n-j} \binom{m}{k-j} p^{k-j} q^{m-k+j} \\ &= p^k q^{n+m-k} \sum_{j=0}^k \binom{n}{j} \binom{m}{k-j} \\ &= \binom{n+m}{k} p^k q^{n+m-k} \quad (\text{Vandermonde's identity}) \\ &\Rightarrow X + Y \sim \text{Bin}(n + m, p) \end{aligned}$$

9 Hypergeometric Distribution, Independence, Expectation

Example: 5-card hand, X = number of aces.

$$P(X = k) = \frac{\binom{4}{k} \binom{48}{5-k}}{\binom{52}{5}}, \quad k = 0, 1, 2, 3, 4$$

General Hypergeometric Distribution: b black marbles, w white marbles; pick a random sample of n marbles (without replacement). Let X = number of white marbles in the sample.

$$P(X = k) = \frac{\binom{w}{k} \binom{b}{n-k}}{\binom{w+b}{n}}$$

Independence of random variables: X, Y are independent if

$$P(X \leq x, Y \leq y) = P(X \leq x) P(Y \leq y) \quad \text{for all } x, y$$

Expectation: $E(X) = \sum_x x P(X = x)$.

Linearity of expectation: $E(X + Y) = E(X) + E(Y)$ — true even if X, Y are dependent. Also $E(cX) = cE(X)$.

Fundamental Bridge: Let $X = \mathbf{1}_A$ (indicator r.v., 1 if event A occurs, else 0). Then $E(X) = P(A)$.

Bernoulli: $X \sim \text{Bern}(p)$: $E(X) = p$.

Binomial — two derivations of $E(X) = np$:

Direct:

$$\begin{aligned} E(X) &= \sum_{k=0}^n k \binom{n}{k} p^k q^{n-k} \\ &= np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j q^{n-1-j} \\ &= np(p+q)^{n-1} = np \end{aligned}$$

Via linearity & indicators:

$$\begin{aligned} X &= X_1 + \cdots + X_n \\ E(X) &= E(X_1) + \cdots + E(X_n) = np \end{aligned}$$

Example: 5-card hand, X = number of aces, X_i = indicator that the i -th card is an ace.

$$E(X) = E(X_1) + \cdots + E(X_5) = 5 \cdot \frac{4}{52} = \frac{5}{13}$$

10 Geometric, Negative Binomial, and First-Success Distributions

Geometric distribution, $\text{Geom}(p)$: independent $\text{Bern}(p)$ trials; X = number of **failures** before the first success.

$$H, TH, TTH, TTTH, \dots$$

$$P(X = k) = q^k p, \quad k = 0, 1, 2, \dots, \quad q = 1 - p$$

Validity: $\sum_{k=0}^{\infty} q^k p = \frac{p}{1 - q} = 1.$

Expectation (let $c = E(X)$): conditioning on the first trial,

$$\begin{aligned} c &= 0 \cdot p + q(1 + c) \\ \Rightarrow c &= \frac{q}{p} \end{aligned}$$

First-Success distribution, $\text{FS}(p)$: X = number of trials until (and including) the first success. Let $Y = X - 1$, so $Y \sim \text{Geom}(p)$:

$$E(X) = E(Y) + 1 = \frac{q}{p} + 1 = \frac{1}{p}$$

Negative Binomial Distribution, parameters r and p : independent $\text{Bern}(p)$ trials; X = number of failures before the r -th success.

PMF: $P(X = n) = \binom{n + r - 1}{r - 1} p^r q^n$, for $n = 0, 1, 2, \dots$

Expectation: $X = X_1 + \dots + X_r$ where X_j = number of failures between the $(j - 1)$ -th and j -th success, each $X_j \sim \text{Geom}(p)$ i.i.d.:

$$E(X) = E(X_1) + \dots + E(X_r) = \frac{rq}{p}$$

Example: Random permutation of $1, 2, \dots, n$. Find the expected number of **local maxima** (a position greater than both its neighbors).

$$\textcircled{3} \quad 2 \quad 1 \quad 4 \quad \textcircled{7} \quad 6 \quad 5$$

(circled values are local maxima)

Let I_j be the indicator that position j is a local maximum.

- Endpoints ($j = 1$ or $j = n$): $E(I_j) = \frac{1}{2}$
- Interior positions: $E(I_j) = \frac{1}{3}$

$$E(\#\text{local maxima}) = \frac{1}{2} + (n-2) \cdot \frac{1}{3} + \frac{1}{2} = \frac{n+1}{3}$$

St. Petersburg Paradox: X = number of coin flips until the coin lands Heads. You receive $\$2^X$. Let $Y = 2^X$. Find $E(Y)$.

11 The Poisson Distribution

St. Petersburg Paradox (resolved):

$$E(Y) = \sum_{k=1}^{\infty} 2^k \cdot P(X = k) = \sum_{k=1}^{\infty} 2^k \left(\frac{1}{2}\right)^k = \sum_{k=1}^{\infty} 1 = \infty$$

The expected payout is **infinite**.

Poisson PMF: $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$, $k = 0, 1, 2, \dots$, where $\lambda > 0$ is the rate parameter.

Validity: $\sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1.$

Expectation:

$$\begin{aligned} E(X) &= \sum_{k=0}^{\infty} k \frac{e^{-\lambda} \lambda^k}{k!} \\ &= e^{-\lambda} \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \\ &= e^{-\lambda} \lambda \cdot e^{\lambda} = \lambda \end{aligned}$$

Often used to model counting successes where there is a large number of trials, each with a small probability of success, e.g.: (1) number of emails received in an hour, (2) number of earthquakes in a region per year.

Poisson Paradigm (Poisson Approximation): Events $A_1, \dots, A_n$ with $P(A_j) = p_j$, n large, p_j small, events independent or weakly dependent. Then the number of A_j 's that occur is approximately $\text{Pois}(\lambda)$, where $\lambda = \sum_j p_j$.

Derivation (Binomial $\rightarrow$ Poisson): Let $n \rightarrow \infty$, $p \rightarrow 0$, with $\lambda = np$ held constant:

$$\begin{aligned} P(X = k) &= \binom{n}{k} p^k (1-p)^{n-k} \\ &\rightarrow \frac{e^{-\lambda} \lambda^k}{k!} \end{aligned}$$

Example: With n people, find the approximate probability that three people share a birthday. There are $\binom{n}{3}$ triplets; for each triplet, the probability of a triple-match is $(1/365)^2$. Let X = number of triplet matches, $X \approx \text{Pois}(\lambda)$ where

$$\lambda = \binom{n}{3} / 365^2$$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-\lambda}$$

12 Continuous Distributions & the Uniform Distribution

	Discrete (X)	Continuous (X)
PMF/PDF	$P(X = x)$	PDF $f_X(x)$, $[P(X = x) = 0]$
CDF	$F(x) = P(X \leq x)$	$F_X(x) = P(X \leq x)$
Expectation	$E(X) = \sum x P(x)$	$E(X) = \int x f_X(x) dx$

Definition (PDF): A r.v. X has PDF $f(x)$ if $P(a \leq X \leq b) = \int_a^b f(x) dx$ for all a, b .

Validity: $f(x) \geq 0$, $\int_{-\infty}^{\infty} f(x) dx = 1$.

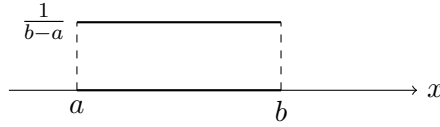
If X has PDF f , the CDF is $F_X(x) = \int_{-\infty}^x f(t) dt$. Conversely, if X has CDF F and X is continuous, $f(x) = F'(x)$.

Fundamental Theorem of Calculus: $P(a < X \leq b) = F(b) - F(a)$.

Variance:

$$\text{Var}(X) = E[(X - E(X))^2] = E(X^2) - [E(X)]^2, \quad \text{SD}(X) = \sqrt{\text{Var}(X)}$$

Uniform Distribution, Unif(a, b): a “completely random” point on $[a, b]$.



$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

$$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x > b \end{cases}$$

$$E(X) = \frac{a+b}{2}$$

LOTUS (Law of the Unconscious Statistician):

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Example: $U \sim \text{Unif}(0, 1)$:

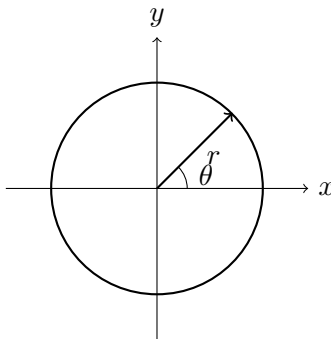
$$E(U) = \frac{1}{2}, \quad E(U^2) = \frac{1}{3}$$

$$\text{Var}(U) = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

13 The Normal (Gaussian) Distribution

Central Limit Theorem: the sum of lots of i.i.d. random variables looks approximately normal.

Standard normal, $Z \sim N(0, 1)$, has PDF $f(z) = c e^{-z^2/2}$. Find the normalizing constant c via the classic polar-coordinates trick:



$$\begin{aligned} \left(\int_{-\infty}^{\infty} e^{-z^2/2} dz \right)^2 &= \iint e^{-(x^2+y^2)/2} dx dy \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r dr d\theta \end{aligned}$$

Let $u = r^2/2$, $du = r dr$:

$$= \int_0^{2\pi} \left[\int_0^{\infty} e^{-u} du \right] d\theta = \int_0^{2\pi} 1 d\theta = 2\pi$$

So $\int_{-\infty}^{\infty} e^{-z^2/2} dz = \sqrt{2\pi}$, hence $c = \frac{1}{\sqrt{2\pi}}$:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

Mean: $E(Z) = 0$ by symmetry (odd function).

Variance: Using integration by parts ($u = z$, $dv = z e^{-z^2/2} dz$):

$$\begin{aligned} E(Z^2) &= \frac{1}{\sqrt{2\pi}} \int z^2 e^{-z^2/2} dz \\ &= \frac{1}{\sqrt{2\pi}} \left[-z e^{-z^2/2} \Big|_{-\infty}^{\infty} + \int e^{-z^2/2} dz \right] \\ &= 0 + 1 = 1 \end{aligned}$$

So $\text{Var}(Z) = 1$.

Notation: $\Phi(z)$ is the standard normal CDF: $\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt$.

Location & Scale: Let $X = \mu + \sigma Z$, $\mu \in \mathbb{R}$ (mean/location), $\sigma > 0$ (SD/scale). Then $X \sim N(\mu, \sigma^2)$.

$$E(X) = \mu, \quad \text{Var}(X) = \sigma^2$$

using $\text{Var}(X + c) = \text{Var}(X)$, $\text{Var}(cX) = c^2 \text{Var}(X)$, and $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ if X, Y independent.

Finding the PDF of $X \sim N(\mu, \sigma^2)$: standardize via $Z = \frac{X - \mu}{\sigma}$:

$$F_X(x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

Differentiating gives the general normal PDF:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

14 Variance (continued) & the Exponential Distribution

Variance of $X \sim \text{Pois}(\lambda)$:

$$E(X^2) = \lambda^2 + \lambda \quad (\text{via manipulating the power series})$$

$$\text{Var}(X) = (\lambda^2 + \lambda) - \lambda^2 = \lambda$$

Variance of $X \sim \text{Bin}(n, p)$: write $X = I_1 + I_2 + \cdots + I_n$, $I_i \sim \text{Bern}(p)$ i.i.d.

$$X^2 = \sum_i I_i^2 + 2 \sum_{i < j} I_i I_j$$

$$E(X^2) = np + n(n-1)p^2$$

$$\text{Var}(X) = np + n(n-1)p^2 - (np)^2 = np(1-p) = npq$$

Example: Emails arrive in time t as $N_t \sim \text{Pois}(\lambda t)$. Find the distribution of T = time until the first email.

$$P(T > t) = P(N_t = 0) = e^{-\lambda t}$$

$$F(t) = 1 - e^{-\lambda t}, \quad f(t) = \lambda e^{-\lambda t}$$

Exponential Distribution, $X \sim \text{Expo}(\lambda)$:

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0, \quad F(x) = 1 - e^{-\lambda x}, \quad x > 0$$

Scaling property: If $X \sim \text{Expo}(\lambda)$, then $Y = \lambda X \sim \text{Expo}(1)$.

Moments of $Y \sim \text{Expo}(1)$:

$$E(Y) = 1, \quad E(Y^2) = 2, \quad \text{Var}(Y) = 1$$

Since $X = Y/\lambda$:

$$E(X) = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

Memoryless Property:

$$P(X > s+t \mid X > s) = P(X > t), \quad \forall s, t \geq 0$$

Proof:

$$P(X > s+t \mid X > s) = \frac{P(X > s+t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$$

15 Moment Generating Functions (MGF)

Memoryless property — corollary: $X \sim \text{Expo}(\lambda)$:

$$E(X \mid X > a) = a + \frac{1}{\lambda}$$

Definition (MGF): A r.v. X has moment generating function $M(t) = E(e^{tX})$ as a function of t , if this is finite on $(-a, a)$ for some $a > 0$.

Why study the MGF?

$$E(e^{tX}) = \sum_{n=0}^{\infty} \frac{E(X^n)}{n!} t^n$$

1. The n -th moment $E(X^n)$ is the coefficient of $\frac{t^n}{n!}$; equivalently $M^{(n)}(0) = E(X^n)$.
2. The MGF determines the distribution — i.e., if X, Y have the same MGF, they have the same PDF/CDF.
3. If X, Y are independent, the MGF of $X + Y$ is $M_{X+Y}(t) = M_X(t) M_Y(t)$.

Examples:

$$X \sim \text{Bern}(p) : M(t) = q + pe^t$$

$$X \sim \text{Bin}(n, p) : M(t) = (pe^t + q)^n$$

$$Z \sim N(0, 1) : M(t) = e^{t^2/2}$$

$$X \sim \text{Expo}(\lambda) : M(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda$$

$$X \sim \text{Pois}(\lambda) : M(t) = e^{\lambda(e^t - 1)}$$

Sum of independent Poissons: $X \sim \text{Pois}(\lambda)$, $Y \sim \text{Pois}(\mu)$, independent:

$$M_{X+Y}(t) = e^{\lambda(e^t - 1)} e^{\mu(e^t - 1)} = e^{(\lambda + \mu)(e^t - 1)}$$

$$\Rightarrow X + Y \sim \text{Pois}(\lambda + \mu)$$

16 Joint, Marginal & Conditional Distributions

Joint CDF: $F(x, y) = P(X \leq x, Y \leq y)$.

Joint PMF (discrete): $P(X = x, Y = y)$.

Marginal distribution: obtained by summing/integrating out the other variable.

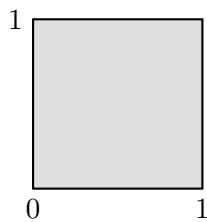
Joint PDF (continuous): $f(x, y)$ such that $P((X, Y) \in B) = \iint_B f(x, y) dx dy$.

Independence: X, Y are independent iff $F(x, y) = F_X(x)F_Y(y)$ for all x, y .

Getting marginals:

$$P(X = x) = \sum_y P(X = x, Y = y) \quad (\text{discrete}), \quad f_X(x) = \int f(x, y) dy \quad (\text{continuous})$$

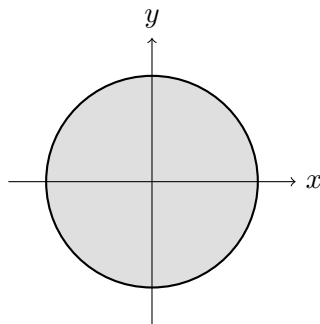
Example — Uniform on the unit square $\{(x, y) : x, y \in [0, 1]\}$:



$$f(x, y) = \begin{cases} 1, & (x, y) \in [0, 1] \times [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

Marginals: $f_X(x) = 1$ for $0 \leq x \leq 1$, similarly $Y \sim \text{Unif}(0, 1)$. Since $f(x, y) = f_X(x)f_Y(y)$, X and Y are independent.

Example — Uniform on the unit disk $\{(x, y) : x^2 + y^2 \leq 1\}$:



$$f(x, y) = \begin{cases} \frac{1}{\pi}, & x^2 + y^2 \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Marginal of X :

$$f_X(x) = \frac{2\sqrt{1-x^2}}{\pi}, \quad -1 \leq x \leq 1$$

Here $f(x, y) \neq f_X(x)f_Y(y)$, so X and Y are NOT independent.

Conditional PDF of Y given X :

$$f_{Y|X}(y | x) = \frac{f(x, y)}{f_X(x)}$$

For the uniform disk: $Y | X = x \sim \text{Unif}(-\sqrt{1-x^2}, \sqrt{1-x^2})$.

17 Order Statistics & the Multinomial Distribution

Example: $X, Y \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$. Find $E|X - Y|$.

By LOTUS:

$$\begin{aligned} E|X - Y| &= \iint_{[0,1]^2} |x - y| dx dy \\ &= \int_0^1 \int_0^x (x - y) dy dx + \int_0^1 \int_0^y (y - x) dx dy \\ &= 2 \int_0^1 \frac{x^2}{2} dx = \frac{1}{3} \end{aligned}$$

Let $M = \max(X, Y)$, $L = \min(X, Y)$. Then $|X - Y| = M - L$, so $E(M) - E(L) = \frac{1}{3}$. Also $M + L = X + Y$, so $E(M) + E(L) = 1$. Solving:

$$E(M) = \frac{2}{3}, \quad E(L) = \frac{1}{3}$$

Example: Find $E|Z_1 - Z_2|$ where $Z_1, Z_2 \stackrel{\text{iid}}{\sim} N(0, 1)$.

Since Z_1, Z_2 are independent, $Z_1 - Z_2 \sim N(0, 2)$, i.e., $Z_1 - Z_2 = \sqrt{2} Z$ for $Z \sim N(0, 1)$.

$$\begin{aligned} E|Z_1 - Z_2| &= \sqrt{2} E|Z| \\ &= \sqrt{2} \cdot \sqrt{\frac{2}{\pi}} \\ &= \frac{2}{\sqrt{\pi}} \end{aligned}$$

Multinomial Distribution:

Definition (story): $\text{Mult}(n, \vec{p})$, $\vec{p} = (p_1, \dots, p_k)$ a probability vector ($p_j \geq 0$, $\sum_j p_j = 1$). Suppose we have n independent trials, each falling into one of k categories with $P(\text{category } j) = p_j$. Let X_j = number of trials landing in category j . Then $\vec{X} = (X_1, \dots, X_k) \sim \text{Mult}(n, \vec{p})$.

Joint PMF:

$$P(X_1 = n_1, \dots, X_k = n_k) = \frac{n!}{n_1! n_2! \dots n_k!} p_1^{n_1} p_2^{n_2} \dots p_k^{n_k}, \quad n_1 + \dots + n_k = n$$

Marginal distribution: Each component is marginally binomial:

$$X_j \sim \text{Bin}(n, p_j), \quad E(X_j) = np_j, \quad \text{Var}(X_j) = np_j(1 - p_j)$$